Proofs Answers

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This is a document for proving the theorems that are listed in the textbook.

2.3

$(a+b)(a-b) = a^2 - b^2$

$$\begin{aligned} & (a+b)(a-b) \\ = \; & (a+b)a - (a+b)b \\ = \; & a^2 + ba - ab - b^2 \\ = \; & a^2 - b^2 \end{aligned}$$

$(a+b)^2 = a^2 + 2ab + b^2$

$$\begin{aligned} & (a+b)^2 = (a+b)(a+b) \\ = \; & (a+b)a + (a+b)b \\ = \; & a^2 + ab + ab + b^2 \\ = \; & a^2 + 2ab + b^2 \end{aligned}$$

$(a-b)^2 = a^2 - 2ab + b^2$

$$\begin{aligned} & (a-b)^2 = (a-b)(a-b) \\ = \; & (a-b)a - (a-b)b \\ = \; & a^2 - ab - ab + b^2 \\ = \; & a^2 - 2ab + b^2 \end{aligned}$$

$(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$

$$\begin{aligned} & (a+b)^3 = (a+b)(a+b)(a+b) \\ = \; & (a^2 + 2ab + b^2)(a+b) \\ = \; & (a^2 + 2ab + b^2)a + (a^2 + 2ab + b^2)b \\ = \; & a^3 + 2a^2b + ab^2 + a^2b + 2ab^2 + b^3 \\ = \; & a^3 + 3a^2b + 3ab^2 + b^3 \end{aligned}$$

$(a-b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$

$$\begin{aligned} & (a-b)^3 = (a-b)(a-b)(a-b) \\ = \; & (a^2 - 2ab + b^2)(a-b) \\ = \; & (a^2 - 2ab + b^2)a - (a^2 - 2ab + b^2)b \\ = \; & a^3 - 2a^2b + ab^2 - a^2b + 2ab^2 - b^3 \\ = \; & a^3 - 3a^2b + 3ab^2 - b^3 \end{aligned}$$

2.4

$a^3 + b^3 = (a+b)(a^2 - ab + b^2)$

$$\begin{aligned} & (a+b)(a^2 - ab + b^2) \\ = \; & (a^2 - ab + b^2)a + (a^2 - ab + b^2)b \\ = \; & a^3 - a^2b + ab^2 + a^2b - ab^2 + b^3 \\ = \; & a^3 + b^3 \end{aligned}$$

$a^3 - b^3 = (a-b)(a^2 + ab + b^2)$

$$\begin{aligned} & (a-b)(a^2 + ab + b^2) \\ = \; & (a^2 + ab + b^2)a - (a^2 + ab + b^2)b \\ = \; & a^3 + a^2b + b^2a - a^2b - ab^2 - b^3 \\ = \; & a^3 - b^3 \end{aligned}$$

$ra + rb + sa + sb = (r+s)(a+b)$

$$ra + rb + sa + sb = r(a+b) + s(a+b) = (r+s)(a+b)$$

2.5

Polynomial Long Division

We consider a polynomial $f(x)$. It can be equally written as

$$a_n x^n + a_n - 1x^{n-1} + ... + a_0 x^0$$

where n is the degree of $f(x)$. We suppose that $f(x)$ is being divided by some $d(x)$. This can be expanded as

$$b_m x^m + b_m x^{m-1} + ... + b_0 x^0$$

where m is the degree of $d(x)$. Define the set of polynomials

$$p_1(x), p_2(x), ..., p_j(x)$$

in the following way:

$$p_1(x) = f(x) - d(x)(\frac{a_n x^n}{b_m x^m})$$

$$p_2(x) = p_1(x) - d(x)(\frac{t}{b_m x^m})$$

where t is the leading term of $p_1(x)$; and, in general (when $i > 1$),

$$p_i(x) = p_{i-1}(x) - d(x)(\frac{t_{i-1}}{b_m x^m})$$

where t_i is the leading term of p_{i-1}, t_0 is $a_n x^n$, and $p_0 - f(x)$, and $i > 1$. The last term, $p_j(x)$, is such that the degree of $p_j(x)$ is less than m. We are to show that

$$\frac{f(x)}{d(x)} = \frac{t_1}{b_m x^m} + \frac{t_2}{b_m x^m} + ... + \frac{t_{j-1}}{b_m x^m} + \frac{p_j(x)}{d(x)}$$

that is,

$$f(x) = \frac{t_1 d(x)}{b_m x^m} + \frac{t_2 d(x)}{b_m x^m} + ... + \frac{t_{j-1} d(x)}{b_m x^m} + p_j(x)$$

Because we defined every $p_i(x)$ such that $p_i(x) = p_{i-1}(x) - \frac{t_{i-1} d(x)}{b_m x^m}$, it follows that $p_j(x) + \frac{t_{j-1} d(x)}{b_m x^m} = p_{j-1}$, and, in general, $p_i(x) + \frac{t_{i-1} d(x)}{b_m x^m} = p_{i-1}$. But this is the process being done in the above sum, but for every i. Thus,

$$\frac{t_1 d(x)}{b_m x^m} + \frac{t_2 d(x)}{b_m x^m} + ... + \frac{t_{j-1} d(x)}{b_m x^m} + p_j(x) = p_1(x) + \frac{t_1 d(x)}{b_m x^m} = f(x)$$

thus completing the proof.

Remainder Theorem

We consider a polynomial $f(x)$ being divided by $x - k$. We are to show that the remainder is $r = f(k)$. Well, by the above result:

$$f(x) = \frac{t_1 d(x)}{x} + \frac{t_2 d(x)}{x} + ... + \frac{t_{j-1} d(x)}{x} + p_j(x)$$

where j is defined from the above algorithm. Because we are dividing by a linear polynomial, the remainder will be a constant term. In particular, this constant will be equal to $p_j(x)$. Now, consider the following sequence $g_1, g_2, ..., g_n$ of terms:

$$g_1 = a_{n-1} x^{n-1} + \frac{a_n x^n}{x} k$$

$$g_2 = a_{n-2}x^{n-2} + \frac{g_1}{x}k$$

and, in general:

$$g_i = a_{n-i}x^{n-i} + \frac{g_{i-1}}{x}k$$

where n is the degree of $f(x)$ and every a_i is defined from the polynomial long division algorithm. Because, as stated, the remainder of $\frac{f(x)}{x-k}$ is $p_j(x)$, we can see that $g_j = p_j(x)$. Now, g_1 consists of adding one a_{n-1}, and a_n k times. g_2 consists of adding one a_{n-2}, a_{n-1} k times, and a_n k^2 times. In general, g_i consists of a_{n-i} 1 time, $a_{n-(i-1)}$ k times, and in general, every a_m $k^{m-(n-i)}$ times, all multiplied by x^{n-i}. Of course, the last term, g_n, is such that $x^{n-i} = 1$, because $i = n$. Thus, we can see that the last term, which is equal to $p_j(x)$, the remainder, is just $a_nk^n + a_{n-1}k^{n-1} + ... + a_0k^0$, that is, $f(k)$, thus completing the proof.

Synthetic Division

Here, we are trying to show that the last result obtained in synthetic division is the remainder, while the other numbers on the bottom line are the respective coefficients of the quotient, if $f(x)$ is being divided by some $x - k$. By the Remainder Theorem, the last result that the last number is the remainder is obtained, because this last number is just $f(k)$, which has been shown to be the remainder of $\frac{f(x)}{x-k}$. As for the other numbers, we already know that the coefficients of the quotient are just all of the $\frac{t_i}{x}$. We can see that the first result of synthetic substitution is just a_nx^{n-1}, which happens to be equal to t_0, that is, $\frac{a_nx^n}{x}$. In the second case, we can see that adding $a_n \times k$ to a_{n-1} to get $(a_{n-1} + a_nk)x^{n-2}$ is just the same as $\frac{t_1}{x}$, that is,

$$\frac{a_{n-1}x^{n-1} + \frac{a_nkx^n}{x}}{x} = (a_{n-1} + a_nk)x^{n-2}$$

and, in general, every ith result from synthetic substitution is just adding the result from the previous substitution, call it s, multiplying it by k, and adding this to a_{n-i}. Thus every result is just

$$(sk + a_{n-i})x^{n-(i+1)}$$

where $i \geq 0$, and the previous term when $i = 0$ is just 1. This is, of course, is equal to

$$\frac{t_{i+1}}{x}$$

that is,

$$(sk + a_{n-i})x^{n-(i+1)}$$

if we let s be the first term of p_{i-1} (because we only add anything to a_{n-i} once because this is a linear term we are dividing by).

Factor Theorem

We want to show that a polynomial $f(x)$ has a factor $x-k$ iff $f(k)=0$. On the one hand, suppose $f(x)$ has a factor $x-k$. Hence,

$$\begin{aligned} & f(x) = (x-k)y \\ \rightarrow \quad & \tfrac{f(x)}{x-k} = y \end{aligned}$$

meaning $\frac{f(x)}{x-k}$ has a remainder of 0, with a quotient of y. On the other hand, suppose that $f(k)=0$. Well, by the Remainder Theorem, $\frac{f(x)}{x-k}$ has a remainder of $f(k)$. Because we assumed $f(k)=0$, this means $\frac{f(x)}{x-k}$ has a remainder of 0, thus meaning,

$$\frac{f(x)}{x-k} = y$$

that is,

$$f(x) = (x-k)y$$

meaning $x-k$ is a factor of $f(x)$.

2.6

Lemma 1

Prove that if $a|b$ and $a|c$, then $a|b+c$.
Suppose $a|b, a|c$. Thus:

$$b+c = ak+aj = a(k+j)$$

for some k, j, completing the proof.

Euclid's Lemma (for review)

We wish to prove

$$\text{if } a, n \text{ are coprime, and } mn = ab \text{ then } n|b$$

We make a complete induction assumption: for all numbers $< ab$, the hypothesis holds for every m, n.
Case 1: $a = n$:
This means $n = 1$ (because $\gcd(a, n) = 1$) and thus $n|b$.
Case 2: $n < a$:

$$mn = ab$$
$$mn - nb = ab - nb$$
$$n(m - b) = b(a - n)$$

We know that $\gcd(n, a - n) = 1$ by Lemma 1 and the assumed coprimality of a, n. Because $\gcd(n, a - n) = 1$, by the induction assumption, $n|b$.
Case 3: $n > a$:

$$mn = ab$$
$$mn - am = ab - am$$
$$m(n - a) = a(b - m)$$

By Lemma 1 and the assumed coprimality of a, n, $\gcd(a, n - a) = 1$. Thus $a|m$ and $aj = m$ for some j:

$$aj = m$$
$$ajn = mn$$
$$ajn = mn = ab$$
$$ajn = ab$$
$$jn = b$$

that is, $n|b$.

Lemma 2

Prove that if a, b are coprimes, then a, b^n are coprimes for every n.
Suppose that a, b^n are not coprime, that is, for some c:

$$cd = a \wedge kc = b^n$$

meaning:

$$kc = b \times b^{n-1}$$

Because a, b are coprimes, and $c \mid a$, c, b are coprimes (becuase, if some $qr = c$ and $qo = b$ for some $q \neq 1$, $qrd = a$, contrary to the assumption of coprimality between a, b). Thus, $c \mid b^{n-1}$ by Euclid's Lemma, meaning, for some h:

$$hc = b \times b^{n-2}$$

Again, because a, b are coprimes, and $c \mid a$, c, b are coprimes. We then repeat this process, but it must end eventually, as n is finite. Thus, we will eventually end up with:

$$cs = b$$

meaning $c|b$, contrary to the assumption that a, b were coprimes.

Rational Roots Theorem

Consider some polynomial

$$a_n x^n + a_{n-1} x^{n-1} + ... + a_1 x + a_0 = 0$$

where every a_i is an integer. We are to show that every rational zero of this polynomial is of the form:

$$\frac{p}{q}$$

where p is a factor of a_0, and q is a factor of a_n. First, we substitute $\frac{p}{q}$ in for x:

$$a_n \frac{p^n}{q^n} + a_{n-1} \frac{p^{n-1}}{q^{n-1}} + ... + a_1 \frac{p}{q} + a_0 = 0$$

$$a_n p^n + a_{n-1} q p^{n-1} + ... + a_1 p q^{n-1} + a_0 q^n = 0 \text{ Multiply both sides by } q^n$$

$$a_n p^n + a_{n-1} q p^{n-1} + ... + a_1 p q^{n-1} = -a_0 q^n$$

$$p(a_n p^{n-1} + a_{n-1} q p^{n-2} + ... + a_1 q^{n-1}) = -a_0 q^n$$

Note that $\frac{p}{q}$ is assumed to be in simplist terms, meaning $\gcd(p, q) = 1$, and consequently, by Lemma 2, $\gcd(p, q^n) = 1$, and thus $\gcd(p, -q^n) = 1$ (for, if some $ab = p$ and $ak = -q^n$ then $-ka = q^n$, meaning both p and q^n share a factor of a, contrary to the assumption of their coprimality). Thus, by Euclid's Lemma, $p|a_0$. On the other hand, we have:

$$a_{n-1} q p^{n-1} + ... + a_1 p q^{n-1} + a_0 q^n = -a_n p^n$$

$$q(a_{n-1} p^{n-1} + ... + a_1 p q^{n-2} + a_0 q^{n-1} = -a_n p^n$$

and with similar reasoning to the first case, we can show that $q|a_n$.